

EXPLORING THE ASSOCIATIONS BETWEEN RESISTANCE TRAINING, INFLAMMATORY MARKERS AND PELVIC PAIN IN ENDOMETRIOSIS: A CROSS-SECTIONAL STUDY

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INTRODUCTION

Endometriosis is a chronic disease that impacts the health of menstruating women and people with a uterus with a wide scope of symptoms ranging from pain during menstruation, fatigue, through to loss of fertility [1]. Endometriosis occurs when the tissue that lines the uterus with each menstrual cycle begins to grow outside of the uterus [1]. These growths are commonly found on the ovaries and fallopian tubes, with severe cases spreading to the bowel, bladder and pelvic cavity [1]. It impacts ~10% of women of reproductive age and while there are several theories regarding the cause of the disease, its pathogenesis remains unclear [2]. The average time taken to reach a diagnosis ranges from 5-12 years, with the most reliable screening tool requiring invasive surgery [3].

A common symptom of endometriosis is pelvic pain, which can have debilitating effects on daily living [1] and has previously been associated with elevated levels of inflammation [4]. Treatment for endometriosis pain is limited and currently consists primarily of hormonal therapy or surgery [5]. Exercise is proven to reduce inflammation in healthy populations [6,7]; furthermore, several studies support the use of exercise to reduce pain, improve quality of life and support mental health in endometriosis [8,9]. However, despite these potential benefits, limited research has investigated the relationship between exercise, inflammation and pelvic pain specifically in endometriosis. This study aims to explore the associations between resistance training, inflammatory markers and pelvic pain in endometriosis.

METHODS

A quantitative cross-sectional study design will be used to explore associations between resistance training, inflammatory markers and pelvic pain. Participants recruited will consist of pre-menopausal women (18-45 years) with a confirmed diagnosis of endometriosis who engage in resistance training. Participants will complete a survey collecting information on resistance training (frequency, intensity, duration), endometriosis (stage, treatment, medication) and pelvic pain (severity, occurrence). The questions relating to exercise patterns will be based on the Muscle-Strengthening Exercise Questionnaire (MSEQ) [10]. Participants will also provide a venipuncture blood sample to be analysed for the inflammatory marker C-reactive protein (CRP). Multivariable regression analyses will be conducted to

examine associations between resistance training behaviours, CRP levels and pelvic pain. Pelvic pain severity will be the primary outcome, with resistance training frequency, duration and intensity included as the primary predictor variables and relevant endometriosis characteristics and participant factors included as covariates.

EXPECED OUTCOMES

After accounting for relevant factors that may contribute to inflammation, such as diet and adipose composition, it is expected there will be an inverse correlation between resistance training volume and pelvic pain severity, alongside a reduction in CRP values. It is possible there may be U-shaped response in which resistance training above a certain volume sees an increase in inflammatory markers and potentially pelvic pain. This could suggest there is an optimal window of exercise to observe benefits in pain reduction.

SIGNIFICANCE

Understanding the associations between resistance training, inflammation and pelvic pain may be significant in informing endometriosis treatment. It could allow practitioners to offer a non-invasive, lifestyle-based alternative for reducing pelvic pain. Further in-depth research is required to understand the contributing factors that may influence pain; however, this study aims to provide a basis for future research investigating the potential role of resistance training in the management of pelvic pain.

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